Light is seen from a point A at a depth d. The Snell's law in product form, equation (6.19), for the refraction happening at the point B on the boundary between the two media is,

$$n_1 \sin i_c = n_2 \sin 90^0 \qquad (6.32)$$

$$n_1 \sin i_c = n_2 \qquad \because \sin 90^0 = 1$$

$$\sin i_c = \frac{n_2}{n_1} \qquad (6.33)$$

From the right angle triangle ΔABC,

$$\sin i_c = \frac{CB}{AB} = \frac{R}{\sqrt{d^2 + R^2}} \qquad (6.34)$$

Equating the above two equations (6.33) and (6.34), $\frac{R}{\sqrt{d^2 + R^2}} = \frac{n_2}{n_1}$

Squaring on both sides, $\frac{R^2}{R^2 + d^2} = \left(\frac{n_2}{n_1}\right)^2$

Taking reciprocal, $\frac{R^2 + d^2}{R^2} = \left(\frac{n_1}{n_2}\right)^2$

Further simplifying,

$$1 + \frac{d^2}{R^2} = \left(\frac{n_1}{n_2}\right)^2; \qquad \frac{d^2}{R^2} = \left(\frac{n_1}{n_2}\right)^2 - 1;$$

$$\frac{d^2}{R^2} = \frac{n_1^2}{n_2^2} - 1 = \frac{n_1^2 - n_2^2}{n_2^2}$$

Again taking reciprocal and rearranging,

$$\frac{R^2}{d^2} = \frac{n_2^2}{n_1^2 - n_2^2}; \qquad R^2 = d^2\left(\frac{n_2^2}{n_1^2 - n_2^2}\right)$$

After taking the squre root, the radius of illumination is,

$$R = d\sqrt{\frac{n_2^2}{\left(n_1^2 - n_2^2\right)}} \qquad (6.35)$$

If the rarer medium outside is air, then, $n_2 = 1$ and we can take $n_1 = n$

$$R = d\left(\frac{1}{\sqrt{n^2 - 1}}\right) \text{ (or) } R = \frac{d}{\sqrt{n^2 - 1}} \qquad (6.36)$$

EXAMPLE 6.9

What is the radius of the illumination when seen above from inside a swimming pool from a depth of 10 m on a sunny day? What is the total angle of view? [Given, refractive index of water is 4/3]

Solution

Given, $n = 4/3$, $d = 10$ m.

Radius of illumination, $R = \frac{d}{\sqrt{n^2 - 1}}$

$$R = \frac{10}{\sqrt{(4/3)^2 - 1}} = \frac{10 \times 3}{\sqrt{16 - 9}}$$

$$R = \frac{30}{\sqrt{7}} = 11.32\,\text{m}$$

To find the critical angle,

$$i_c = \sin^{-1}\left(\frac{1}{n}\right)$$

$$i_c = \sin^{-1}\left(\frac{1}{4/3}\right) = \sin^{-1}\left(\frac{3}{4}\right) = 48.6^\circ$$

The total angle of view of the cone is,

$$2i_c = 2 \times 48.6^\circ = 97.2^\circ$$

6.4.7.5 Optical fibre

Transmitting the signals through optical fibres is possible due to the phenomenon of total internal reflection. **Optical fibres consist of inner part called *core* and outer part called *cladding* (or) *sleeving*.** The refractive index of the core must be higher than that of the cladding for total internal reflection to happen. Signal in the form of light is made to incident inside the core-cladding boundary at an angle greater than

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